

MD TANVIR ISLAM

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EDUCATION

February 2023 February 2025	M.Sc., SUNGKYUNKWAN UNIVERSITY (SKKU) , South Korea Department of Computer Science and Engineering CGPA : 4.19/4.50 Thesis : A Study on Haze Aware Single Image Dehazing for Adverse Atmospheric Conditions.	Transcript
March 2022 February 2023	Korean Language and Cultural Program, PAI CHAI UNIVERSITY , South Korea Korean Language and Cultural Program TOPIK Level : 4/6 As a prerequisite, I completed this program under the Global Korea Scholarship (GKS).	Transcript
January 2015 January 2019	B.Sc., North Western University (NWU) , Bangladesh Department of Computer Science and Engineering CGPA : 3.98/4.00 Thesis : Study on diabetes prediction for typical and non-typical cases using machine learning approaches.	Transcript

RESEARCH EXPERIENCES

BK : BRAIN KOREA | IF : IMPACT FACTOR

March 2023 Present	Research Assistant, VIS2KNOW LAB , Sungkyunkwan University, South Korea <i>Sungkyunkwan University, South Korea</i> ➤ Data curation and research in AI image detection, image dehazing, and low-light image enhancement. ➤ Our research on image dehazing, LLIE, and AI-synthesized image detection has led to three papers accepted and under review by top-tier conferences and journals such as ACM Multimedia (BK IF : 4), Asian Conference on Computer Vision (ACCV) (BK IF : 1), Alexandria Engineering Journal (AEJ) (IF : 6.8, Q1) and three patents under review by the US Patent Office.	Lab Director
July 2018 August 2021	Research Assistant, CSHI-RLAB , North Western University, Bangladesh <i>North Western University, South Korea</i> ➤ Collected real-life datasets to perform research using machine-learning techniques and published several papers at different international conferences, which achieved around 100 citations so far.	Lab Director

TEACHING EXPERIENCES

March 2024 Present	Teaching Assistant, DEPARTMENT OF AI , Sungkyunkwan University, South Korea <i>Sungkyunkwan University, South Korea</i> ➤ Assisted with curriculum planning for courses and held office hours to support and mentor students. ➤ Facilitated group discussions and study sessions to enhance collaborative learning.
January 2019 December 2020	Part-time Lecturer, SOUTH-EAST ENGINEERING COLLEGE , Khulna, Bangladesh <i>Affiliated by University of Rajshahi, Bangladesh</i> ➤ Planned, prepared and presented lectures while assessing student assignments. ➤ Maintained communication with teachers and students.

RESEARCH OUTCOMES

BK : BRAIN KOREA | IF : IMPACT FACTOR | [SEE ALL PAPERS](#)

Journal Papers (3)	Published : IEEE Access (IF : 3.9, Q1) Accepted : AEJ (IF : 6.2, Q1), EAAI (IF : 7.8, Q1)
Conference Papers (13)	Published : ACM MM'24, (Rank : A BK IF : 4), ACCV'24 (Rank : B BK IF : 1.0), MIET'24 (Scopus), ICICC, IJCCI, ICCCNT, ICCISIoT
Patents (4)	UK (1 Published), US Patent (1 passed evaluation round), US Patent (2 in review)
Book Chapter (1)	Rezuana Haque & Md Tanvir Islam (2024). "Neural Dynamics : Unravelling the Complexity of Brain Activity." In Brain Networks in Neuroscience : Personalization Unveiled Via Artificial Intelligence. Chapter 1, pp. 1-19. (In review)

SKILLS AND INTERESTS

Language	Bengali (Native), English (IELTS (2019) : 7.0, no band < 6.5), Korean (TOPIK (2022) : 4/6).
Programming	Python; Framework : PyTorch, Keras; Database : PostgreSQL, MySQL.
Tools	PowerBI, LaTeX, PyCharm, Jupyter, VSCode, Photoshop, Illustrator, Figma, Microsoft Suites.
Interests	Computer Vision, Deep Learning, Machine Learning, Image Processing, Bioinformatics.

PROJETS

HAZESPACE2M – [GITHUB.COM/TANVIRNWU/HAZESPACE2M](https://github.com/TANVIRNWU/HAZESPACE2M)

1ST LEAD AUTHOR

A large-scale hazy dataset of over 2 Million images across 10 distinct levels and propose a novel framework for haze-aware single image dehazing that significantly improves PSNR by 2.41%, SSIM by 17.14%, and MSE by 10.2% over general dehazers.

TRIFUSE & LOLI-STREET – [GITHUB.COM/TANVIRNWU/TRIFUSE](https://github.com/TANVIRNWU/TRIFUSE)

1ST LEAD AUTHOR

First to introduce a 33K paired low-light street scene dataset with object detection labels and propose TriFuse, a transformer-wavelet-diffusion model that achieves SOTA results for real-life low-light image enhancement for autonomous systems.

LIGHTCDC

1ST LEAD AUTHOR

We developed a processed dataset for damaged crop classification and proposed a custom lightweight CNN model with just 1.13M parameters, achieving 92% accuracy, demonstrating its effectiveness for real-life applications.

CHILD

2ND LEAD AUTHOR

We introduce a custom computer vision-based model and a new labeled video dataset for detecting and recognizing hazardous activities performed by children in indoor settings.

Machine Learning

Deep Learning

Computer Vision

OpenCV

Python

PyTorch

Overleaf

PyCharm

LaTeX

Photoshop

Illustrator

TRAININGS

August 2019 September 2019	LICT Top-up IT Training, ICT Division, Bangladesh <i>Bangladesh Computer Council (Certified by George Washington University, Washington, D.C.)</i> ▶ The training includes 240 hours of technical skills such as Java, HTML, CSS, JavaScript, Bootstrap, and MVC Pattern, and 60 hours of other soft skills like presentation and communication.	Certificate
February 2018	Google Bangladesh Android StudyJam, GDG DHAKA, Bangladesh <i>ICT Division, GDG Dhaka, Google Developers</i> ▶ A 225-hour Android app development course with regular mini-project assignments, culminating in a final complete app development project.	Certificate
January 2019 Present	Self Learning, <i>Some of the courses and books I completed to grow my knowledge.</i> ▶ PyTorch for Deep Learning Bootcamp (52 hours) ▶ Python for Data Science, AI & Development (25 hours) ▶ Python for Data Science and Machine Learning Bootcamp (25 hours) ▶ Foundation of User Experience (UX) Design (19 hours) ▶ Modern Computer Vision with PyTorch ▶ Python Deep Learning Projects ▶ Advanced Deep Learning with Python	Online, Books↓ Udemy Mar 2024 IBM Mar 2024 Udemy Sept 2019 Google Oct 2021 Book by V Kishore Ayyadevara Book by Matthew Lamons Book by Ivan Vasilev

AWARDS, SCHOLARSHIPS AND ACTIVITIES

Academic Excellence Award Winner

Nov 2024

The NIIED, Government of South Korea, selected me as one of the three students for this award based on the research outcomes.

BK21 FOUR Project, funded by MOE & NRF, South Korea

No. 519999091384

Our project and accepted paper, HazeSpace2M, is funded by the BK FOUR project, which is funded by the **Ministry of Education (MOE)** and the **National Research Foundation of Korea (NRF)**.

National Research Foundation of Korea (NRF) grant, South Korea

No. 2021R1F1A1050022

One of my papers was funded by the NRF grant funded by the **Ministry of Science and ICT (MSIT)**

16th Korean Language Society Scholarship, Pai Chai University, Republic of Korea

Dec 2023

Global Korea Scholarship (GKS), NIIED, Republic of Korea

Aug 2022 – Present

First position in Merit List, B.Sc. in CSE, North Western University, Bangladesh

Jan 2015 - Jan 2019

Tuition Fee Waiver and Stipend, North Western University, Bangladesh

Dec 2014 - Jan 2019

Research Paper Reviewer, International Conference of Pattern Recognition (ICPR), Kolkata, India.

July 2024

Volunteering to Rescue Animals, Khulna Cat Society, Khulna, Bangladesh

July 2016 - Present

Paper Presentation in ICCISIoT Conference, NIT Agartala, India

Dec 2019

REFERENCES

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